HCMUS_Beyonders at AstroCLIMB: Same Paper or Citing Paper? Linking Astronomy Figures with OCR, Citation Graphs, and LLMs

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0.7421
macro-F1, full test set
0.7329 with Qwen3-8B, the last step
backed by held-out data

01 Four relations between two objects

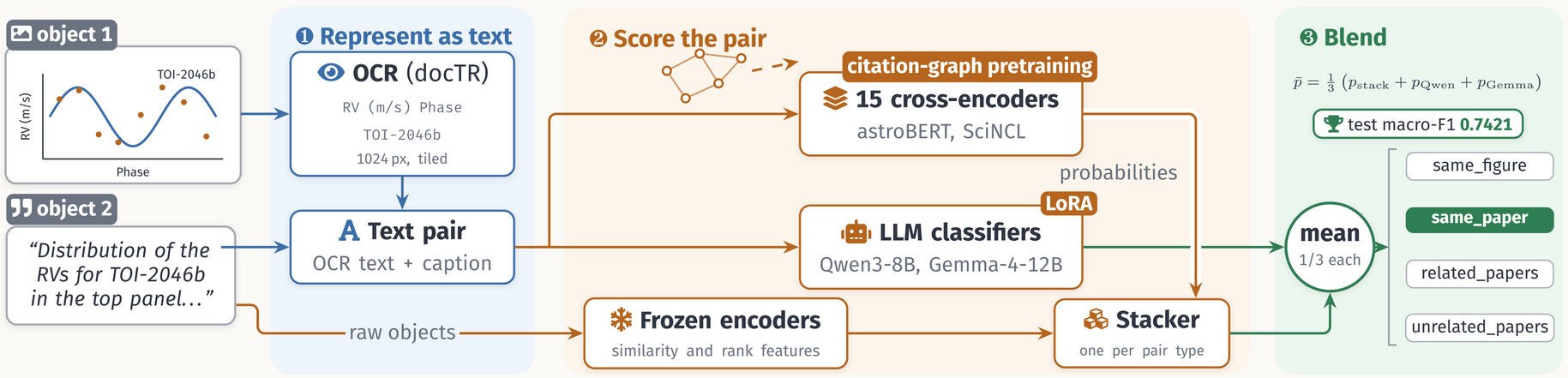


10,000 training and 10,000 test pairs; test papers are unseen; ranked by macro-F1.

02 Three difficulties, three answers

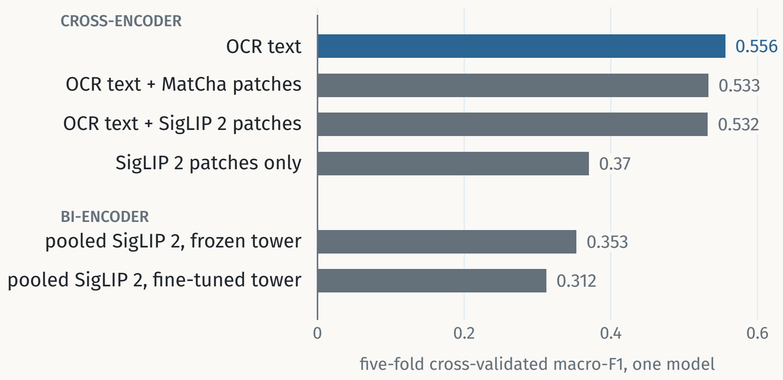
DIFFICULTY	OUR ANSWER
Plots keep their content in axis labels, object names and legends	Read every figure with **OCR**: every pair becomes a pair of texts
Only 10,000 labelled pairs, and the test papers are unseen	**Pretrain** cross-encoders on 600K pairs from the public citation graph
A paper and the papers it cites share a topic	Average in **LoRA-tuned LLMs** that make different errors

03 Read figures as text, score the pair, blend two kinds of model



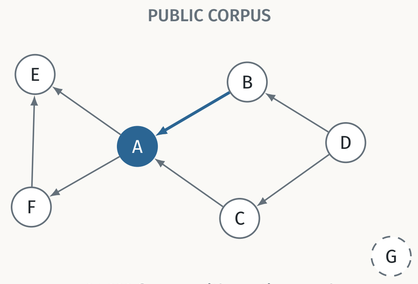
Stage 1 (blue): 15 cross-encoders and a stacker. Stage 2 (orange): Qwen3-8B and Gemma-4-12B with LoRA. Label: argmax of the equal-weight average (green); `same_figure` only for image–caption pairs.

04 OCR text beats every image input we tried



Early system, one model per row. A vision–language LLM reading the raw figure was not tested.

05 600K graph pairs reach unseen papers



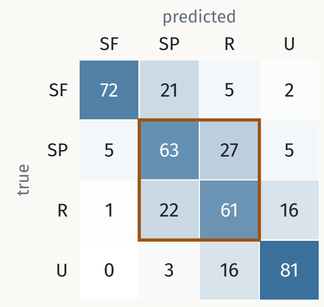
generate

600K LABELLED PAIRS	FROM THE GRAPH
`same_figure`	a figure's OCR text + its caption
`same_paper`	two objects of **A**
`related_papers`	**A** and **B**: linked by a reference
`unrelated_papers`	**A** and **G**: no link half drawn from topically similar papers

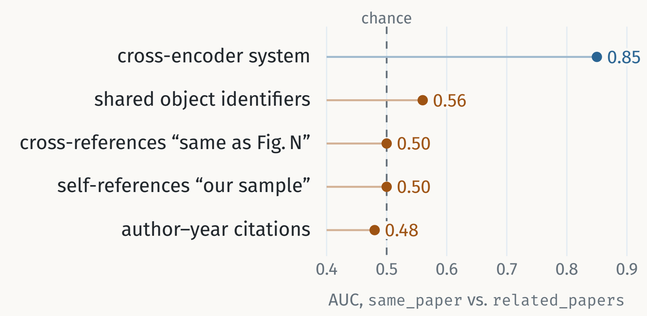
60K per (pair type, label) combination; image objects carry real OCR text

Pretraining: **+0.022** test macro-F1.
Real OCR text in it: a further **+0.038** in cross-validation.

06 Same paper vs. citing paper is the dominant error

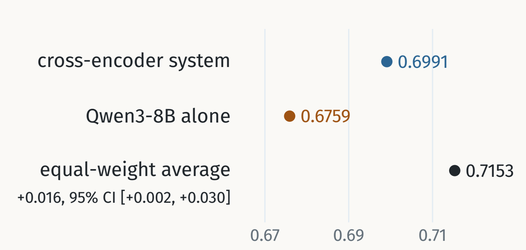


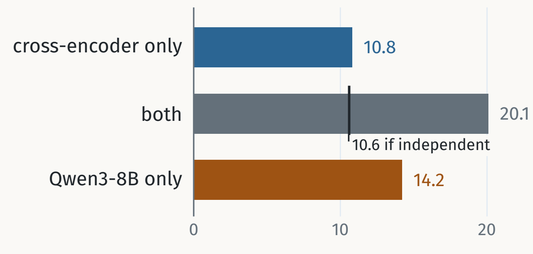
% of true class, 10,000 held-out pairs
SF `same_figure` SP `same_paper`
R `related_papers` U `unrelated_papers`



The boxed confusion covers 14.7% of all pairs and is as common on caption–caption pairs: not a vision problem.

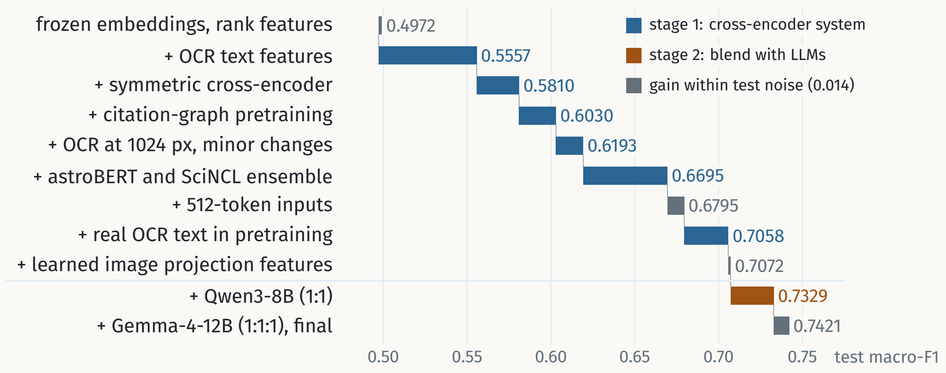
07 The LLM is weaker alone but errs differently





Fold 1. 35% of the cross-encoder's errors and 41% of Qwen3-8B's are not made by the other model.

08 From 0.4972 to 0.7421 on the test leaderboard



Noise threshold 0.014 (bootstrap, 95%). Gemma-4-12B had no held-out validation; its weighting was chosen between two test submissions.

09 What to keep

1. Read astronomy figures as text.
2. Turn the citation graph into training pairs to reach unseen papers.
3. When cues stop helping, blend models that make different errors.

Open question: would a vision–language LLM that reads the raw figure beat OCR text?

`github.com/NLPQuy/WASP_2026`

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